

c. Prime numbers

Definitions

- **Def.** A **prime number** is an integer greater than 1 for which the only divisors are 1 and itself.
- **Ex.** 2,3,5,7...
- We'll now show that the primes are the “building blocks” of positive integers
- i.e., the “Fundamental theorem of arithmetic” says that every integer > 1 can be written uniquely as a product of primes
 - **Simpler thm.** Every natural number > 1 has a prime divisor i.e., $n > 1 \Rightarrow \exists \text{ prime } p \text{ such that } p|n$
 - The way to prove this uses induction, as with many universally quantified statements in number theory...

Induction review!

- **Goal of induction:** we want to prove a universal statement of the form:
 - $\forall n > 0, \phi(n) = 1$.
- **Steps:**
 - Prove the “base case” $\phi(1) = 1$
 - Prove the “inductive step” $\forall n > 0, \phi(n) = 1 \Rightarrow \phi(n + 1) = 1$
- **(recall) Simpler thm.** Every positive integer > 1 has a prime divisor i.e., $n > 1 \Rightarrow \exists \text{ prime } p \text{ such that } p|n$
 - But a totally straightforward inductive approach doesn't work!
 - i.e., in the induction step, we'd need to show $n \text{ has prime divisor} \Rightarrow \exists p|(n + 1)$
 - Consider $n + 1$. Either it's prime or it's not. If it is, we're done (since $(n + 1)|(n + 1)$)
 - But if it's not, then $n + 1 = a \cdot b$ and we'd need to use Inductive Hypothesis (i.e., that a prime divides n) to show that a prime $p|a$ or $p|b$ – this isn't obvious!

Strong induction

- Instead, let's consider “strengthening” inductive hypothesis
- In strong induction we use:
- **Def** (Strong Inductive Principle):
- $(\forall n > 0, (\forall k < n, \phi(k) = 1) \Rightarrow (\phi(n) = 1)) \Rightarrow \forall n > 0, \phi(n) = 1$
 - *Proof by contrapositive*: Suppose RHS is false, i.e., $\exists n \phi(n) = 0$ and so if we look at the set of “counterexamples”: $\{n \mid \neg \phi(n) = 1\}$ is a non-empty set of natural numbers
 - So it must have a minimum element, n_0 . But since that's the minimum counterexample, we must have $\forall k < n_0, \phi(k) = 1$.
 - But notice this contradicts the LHS (since this does not imply that $\phi(n_0) = 1$, because in fact, $\phi(n_0) \neq 1$)

Return to proof of “simpler theorem”

- (Recall) Simpler thm. Every natural number > 1 has a prime divisor
i.e., $n > 1 \Rightarrow \exists \text{ prime } p \text{ such that } p|n$
- **Pf.** Let the statement $\phi(n) = (n > 1) \Rightarrow \exists \text{ prime } p \text{ so that } p|n$
 - Then by Strong Induction, assume $\phi(k)$ is true whenever $k < n$
 - Proceed as before: if n is prime then $n|n$ and we're done
 - So assume n is not prime. Then $n = a \cdot b$ for $1 < a, b < n$
 - By inductive hypothesis, $\exists \text{ prime } p, \text{ so that } p|a$. So this implies $p|n$. QED

Cor. There are infinitely many primes!

- **Pf:** Consider any finite list of prime numbers $p_1, p_2, \dots, p_k$. We'll show it's missing some prime number!
- Let $p = p_1 p_2 \dots p_k$
 - Then consider $n = p + 1 = p_1 \cdot p_2 \cdot \dots \cdot p_k + 1$.
 - Now either n is prime or it's not
 - If it is, then there's at least one prime not on the list!
 - So let's assume n is not prime
 - But then by **simpler theorem**, some prime $p' \mid n$.
 - Claim: p' cannot be in our list! (Why?)
 - Because if it's in our list then $p' \mid p$
 - But since also $p' \mid n$ we know...
 - $p' \mid (n-p)=1$ (why?)
 - But no prime number can divide 1 !!

A helpful lemma

- **Helpful Lemma:** If p is prime and so are $p_1, p_2, \dots, p_k$ (not necessarily distinct), and $p | p_k \cdot p_{k-1} \cdot \dots \cdot p_1$ then p must be p_i for some i
- **Pf (Lemma):** how should we proceed?
 - Standard induction on number of primes, k
 - Base case: $k=1$, then $p | p_1 \Leftrightarrow p = p_1$
 - Inductive Hypothesis: Assume for k , to prove for $k + 1$
 - So consider instance of the $k + 1$ case, i.e., suppose $p | p_1 \cdot p_2 \cdot \dots \cdot p_{k+1}$
 - Then recall that for prime p , if $p | a \cdot b \Rightarrow p | a$ or $p | b$.
 - Now suppose $a = p_1 \cdot p_2 \cdot \dots \cdot p_k$ and $b = p_{k+1}$
 - Then either $p | a$, in which case $p = p_i$ for some $i \in \{1, \dots, k\}$ by inductive hypothesis
 - Or $p | b$, in which case $p | p_{k+1}$ and so $p = p_{k+1}$

Fundamental Theorem of Arithmetic

- Recall **Thm** (FTA): Every $n > 1$ can be written uniquely (up to changing the order) as a product of primes.
- **Pf.** (existence): Easy to prove via strong induction (exercise!)
- **Pf.** (uniqueness):
 - Suppose the prime factorizations aren't unique
 - Let n **be the least** natural number that doesn't have a unique prime factorization
 - So $n = p_1 \cdot \dots \cdot p_k = p'_1 \cdot \dots \cdot p'_{k'}$
 - Now $p_1 | n$ so by **Helpful Lemma**, we must have $p_1 = p'_j$ for some j .
 - But then we can divide both sides by p_1 and obtain a smaller counter-example!
 - This contradicts minimality of n . QED